21-242 Recitation 12

Linear Transformations — Q3. Unique linear map sending a basis to prescribed vectors

Course-level solution, with a later-course alternative when useful

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Prompt

If $\vec{v}_1, \ldots, \vec{v}_n$ are a basis for V and $\vec{w}_1, \ldots, \vec{w}_n \in W$, there exists a unique linear transformation $T : V \to W$ with $T(\vec{v}_i) = \vec{w}_i$ for all $i \in [n]$.

Course-level solution

Existence. Each $\vec{v} \in V$ may be written uniquely as $\vec{v} = \sum_{i=1}^{n} a_i \vec{v}_i$. Define

$$T(\vec{v}) := \sum_{i=1}^{n} a_i \vec{w}_i.$$

If $\vec{v} = \sum a_i \vec{v}_i$ and $\vec{u} = \sum b_i \vec{v}_i$, then $c\vec{v} + d\vec{u} = \sum (ca_i + db_i)\vec{v}_i$, so

$$T(c\vec{v} + d\vec{u}) = \sum (ca_i + db_i)\vec{w}_i = c\,T(\vec{v}) + d\,T(\vec{u}).$$

By construction $T(\vec{v}_i) = \vec{w}_i$.
Uniqueness. If S is linear and $S(\vec{v}_i) = \vec{w}_i$, then for $\vec{v} = \sum a_i \vec{v}_i$,

$$S(\vec{v}) = \sum a_i S(\vec{v}_i) = \sum a_i \vec{w}_i = T(\vec{v}).$$

Later-course solution (using ideas from the rest of the handout)

In matrix language (next section): the unique T is the map whose matrix $[T]_{\mathcal{CB}}$ has jth column $[\vec{w}_j]_{\mathcal{C}}$, for any basis $\mathcal{C}$ of W. Uniqueness is the statement that a linear map is determined by its matrix.